The documentation and process conversion measures necessary to comply with this revision shall be completed by 30 September 1999

INCH-POUND

MIL-PRF-19500/368D
30 June 1999
SUPERSEDING
MIL-S-19500/368C
3 January 1994

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER
TYPES: 2N3439, 2N3439L, 2N3440, AND 2N3440L
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, low-power, high voltage transistors. Four levels of product assurance are provided for each encapsulated device types as specified in MIL-PRF-19500, and two levels of product assurance for each unencapsulated device type die.

1.2 Physical dimensions. See figure 1 (similar to TO- 5 and TO-39) and 2 (JANHC and JANKC).

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Types	P_T 1/ T_A = +25°C	P_T 2/ T_C = +25°C	V_{CBO}	V_{EBO}	V_{CEO}	I_C	T_{STG} and T_{OP}
	W	W	V dc	V dc	V dc	A dc	°C
2N3439	0.8	5.0	450	7	350	1.0	-55 to +200
2N3440	0.8	5.0	300	7	250	1.0	-55 to +200

1/ Derate linearly 4.57 mW/°C for T_A > +25°C.

2/ Derate linearly 28.5 mW/°C for T_C > +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics.

	h_{FE2} 1/ V_{CE} = 10 V dc I_C = 2 mA dc	h_{FE1} 1/ V_{CE} = 10 V dc I_C = 20 mA dc	$\|h_{fe}\|$ V_{CE} = 10 V dc I_C = 10 mA dc f = 5 MHz	C_{obo} V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	$V_{BE(sat)}$ 1/ I_C = 50 mA dc I_B = 4 mA dc	$V_{CE(sat)}$ I_C = 50 mA dc I_B = 4 mA dc
				pF	V dc	V dc
Min	30	40	3			
Max		160	15	10	1.3	0.5

1/ Pulsed, (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified manufacturer's list before contract award (see 4.2 and 6.3).

3.2 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

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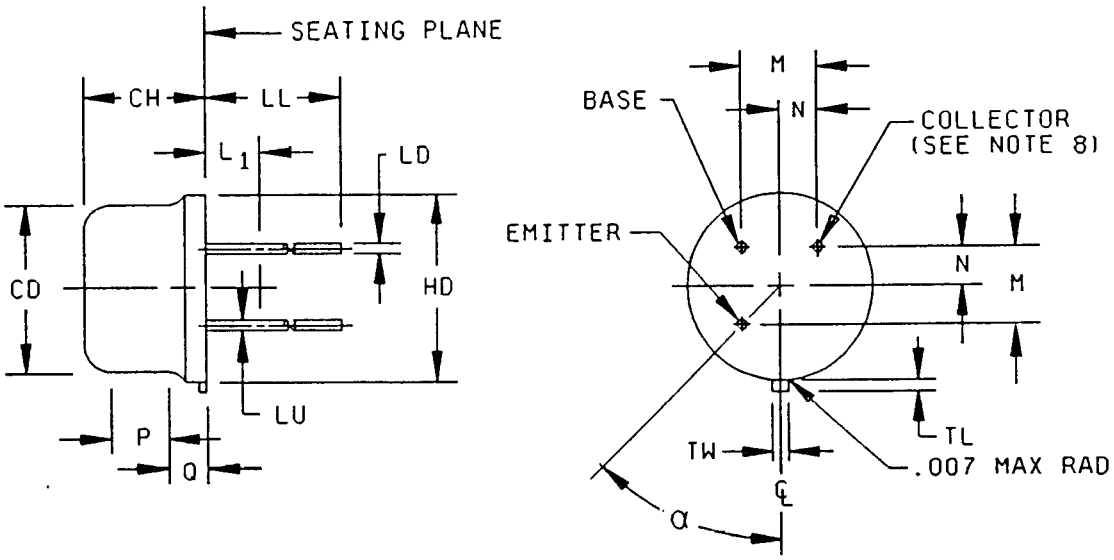


FIGURE 1. Physical dimensions (similar to TO- 5 and TO-39).

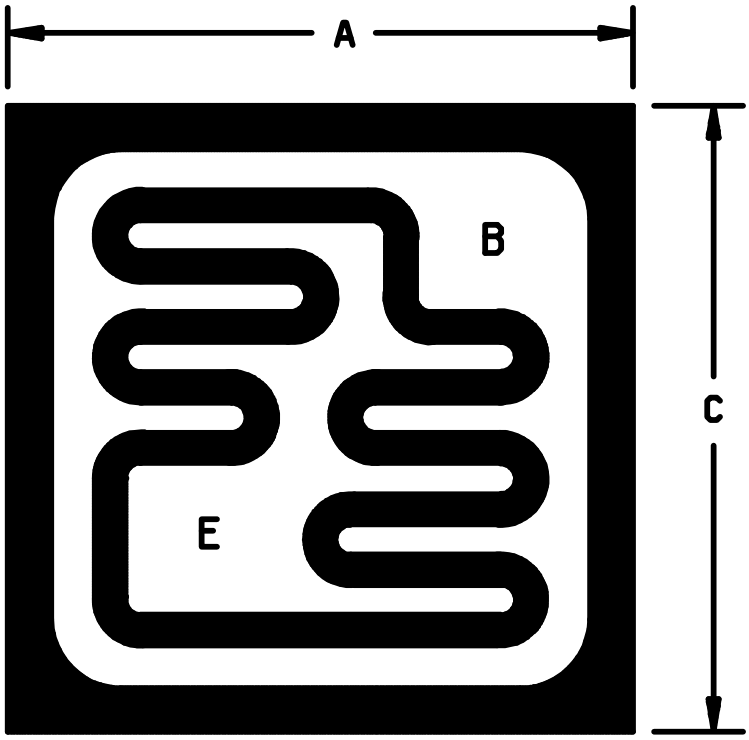
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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	0.305	0.335	7.75	8.51	
CH	0.240	0.260	6.10	6.60	
HD	0.335	0.370	8.51	9.40	
LD	0.016	0.021	0.41	0.53	3, 10
LL	See notes 10, 12, and 13				
L1		0.050		1.27	11
LU	0.016	0.019	0.41	0.48	4, 10
M	0.1414 Nom		3.591 Nom		7
N	0.0707 Nom		1.796 Nom		7
P	0.100		2.54		5
Q					6
TL	0.029	0.045	0.74	1.14	9
TW	0.028	0.034	0.71	0.86	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Measured in the zone beyond 0.250 inch (6.35 mm) from the seating plane.
4. Measured in the zone 0.50 inch (1.27 mm) from the seating plane.
5. Variations on dimension CD in this zone shall not exceed 0.010 inch (0.25 mm).
6. Outline in this zone is not controlled.
7. When measured in a gauging plane 0.054 +0.001, -0.000 inch (1.37 +0.03, - 0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within 0.007 inch (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance.
8. The collector shall be electrically connected to the case.
9. Measured from the maximum diameter of the actual device.
10. All three leads (see 3.4.1).
11. Diameter of leads in this zone is not controlled.
12. For transistor types 2N3439 and 2N3440, L = 0.500 inch (12.70 mm) minimum, and 0.750 inch (19.05 mm) maximum.
13. For transistor types 2N3439L and 2N3440L, L = 1.500 inches (38.10 mm) minimum, and 1.750 inches (44.45 mm)maximum.

FIGURE 1. Physical dimensions - Continued.

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Ltr	Inches		Millimeters	
	Min	Max	Min	Max
A	.038	.044	0.97	1.12
C	.038	.044	0.97	1.12

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die thickness are, 0.006 inch (0.15 mm) to 0.012 inch (0.30 mm).
   Metallization, top: Al = 17,500 Å minimum, 20,000 Å nominal.
   Back: Au = 2,500 Å minimum, 3,000 Å nominal.
   Bonding pad: B = 0.004 inch (0.10 mm) by 0.005 inch (0.13 mm).
   E = 0.004 inch (0.10 mm) by 0.005 inch (0.13 mm).
4. Backside is collector.
5. Requirements in accordance with MIL-PRF-19500 are performed in a TO-5 package.

FIGURE 2. Physical dimensions JANHC and JANKC (die).

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3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, MIL-HDBK-6100 and on figures 1 (similar to TO- 5 and TO-39) and 2 (JANHC and JANKC) herein.

3.4.1 Lead material and finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and herein.

4.3 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with MIL-PRF-19500 (table IV) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
7	Hermetic seal (optional)	1/
9	I_{CBO1} and h_{FE1}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CBO1} ; h_{FE1}; ΔI_{CB01} = 100 percent of initial value or 0.5 µA dc, whichever is greater; Δh_{FE1} = ± 15 percent of initial value.	I_{CBO1} and h_{FE1}
12	See 4.3.2, 240 hours minimum	See 4.3.2, 80 hours minimum
13	Subgroups 2 and 3 of table I herein; ΔI_{CB01} = 100 percent of initial value or 200 nA dc, whichever is greater; Δh_{FE1} = ± 15 percent of initial value.	Subgroup 2 of table I herein; ΔI_{CB01} = 100 percent of initial value or 200 nA dc, whichever is greater; Δh_{FE1} = ± 15 percent of initial value.

1/ Hermetic seal test shall be performed in either screen 7 or screen 14.

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4.3.1 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500. As a minimum, die shall be 100 percent probed to ensure the assembled chips will meet the requirements of group A, subgroup 2.

4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: T_A = Room ambient as defined in the general requirements (4.5) of MIL-STD-750.

V_{CB} = 10 - 30 V dc, P_T = 600 mW.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.3 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, Method 3131.

a. I_M measurement current ---------- 10 mA.

b. I_H forward heating current ---------- 250 mA.

c. t_H heating time ---------- 100 ms.

d. t_{MD} measurement delay time ---------- 30 – 60 µs.

e. V_{CE} collector-emitter voltage ---------- 10 V dc minimum.

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 20 °C/W.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in VIa (JANS) of 4.4.2.1. Electrical measurements (end-points) requirements shall be in accordance with group A, subgroup 2 herein. Delta requirements shall be in accordance with 4.5.3 herein. See 4.4.2.2 herein for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) requirements shall be in accordance with group A, subgroup 2 herein. Delta requirements shall be in accordance with 4.5.3 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 - 30 V dc, 2,000 cycles.
B5	1027	V_{CB} = 10 - 30 V dc; T_A = 160°C max for96 hours, with P_T adjusted according to the chosen T_A to give an average T_J = 275°C.

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4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). 1/

Step	Method	Condition
1	1039	Steady-state life: Test condition B, 340 hours, V_{CB} = 10 -30 V dc, T_J = 150°C min. No heat sink or forced-air cooling on the devices shall be permitted. n = 45 devices, c = 0
2	1039	The steady state life test of step 1 shall be extended to 1,000 hours for each die design. Samples shall be selected from a wafer lot every twelve months of wafer production, however, Group B shall not be required more than once for any single wafer lot. n = 45, c = 0.
3	1032	High-temperature life (non-operating), T_A = +200°C. n = 22, c = 0

1/ Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new "assembly lot" option is exercised, the failed assembly lot shall be scrapped.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection, Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS).and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.3 herein.

4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C6	1026	V_{CB} 10 - 30 V dc; T_J = + 150°C. No heat sink or forced-air cooling on the devices shall be permitted.

4.4.3.2 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C6		Not applicable

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500 and herein. End point electrical measurements, see group A, subgroup 2 and 4.5.3 herein.

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4.4.4.1 Group E inspection, table IX of MIL-PRF-19500.

Subgroup	Method	Condition	Sampling plan
E1	1051	Test condition C, 500 cycles	12 devices, c = 0
E1	1071		12 devices, c = 0
E2	1037	Intermittent operation life: $V_{CB} \geq 10$ V dc , 6,000 cycles.	45 devices, c = 0
E3, E4, E5		Not applicable	

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following conditions shall apply:

a. Collector current magnitude during power application shall be 150 mA dc for $R_{\theta JC}$.

b. Collector emitter voltage magnitude shall be 10 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference temperature measuring point shall be $+25°C \leq T_R \leq +35°C$. The chosen reference temperature shall be recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case for $R_{\theta JC}$.

f. Maximum limit shall be $R_{\theta JC} = 35°C/W$.

4.5.3 Delta requirements. Delta requirements shall be as specified below:

Step	Inspection	MIL-STD-750		Symbol	Unit
		Method	Conditions		
1	Collector-base cutoff current	3036	Bias condition D, V_{CB} 2N3439 = 360 V dc V_{CB} 2N3440 = 250 V dc	ΔI_{CB01} 1/	100 percent of initial value or 200 nA dc, whichever is greater.
2	Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 2.0 mA dc; pulsed see 4.5.2	Δh_{FE1} 1/	±15 percent change from initial reading.

1/ Devices which exceed the group A limits for this test shall not be accepted.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical 3/ examination	2071	n = 45 devices, c = 0				
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to 3/ 4/ 5/ solvent	1022	n = 15 devices, c = 0				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Heremetic seal 4/	1071	n = 22 devices, c = 0				
Fine leak Gross leak						
Electrical measurements 4/		Group A, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs, n = 11 wires, c = 0				
Subgroup 2						
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 7 V dc	I_{EBO1}		10	µA dc
Collector to emitter cutoff	3041	Bias condition D	I_{CEO}		2	µA dc
2N3439 2N3440		V_{CE} = 300 V dc V_{CE} = 200 V dc				
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc	ICEX		5	µA dc
2N3439 2N3440		V_{CE} = 450 V dc V_{CE} = 300 V dc				
Collector to base cutoff current	3036	Bias condition D	I_{CBO1}		2	µA dc
2N3439 2N3440		V_{CB} = 360 V dc V_{CB} = 250 V dc				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 – Continued						
Collector to base cutoff current	3036	Bias condition D	I_{CBO2}		5	µA dc
2N3439 2N3440		V_{CB} = 450 V dc V_{CB} = 300 V dc				
Base emitter voltage (nonsaturated)	3066	Test condition A, pulsed (see 4.5.1), I_C = 50 mA dc, I_B = 4 mA dc	$V_{BE(sat)}$		1.3	V dc
Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1), I_C = 50 mA dc, I_B = 4 mA dc	$V_{CE(sat)}$		0.5	V dc
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), V_{CE} = 10 V dc, I_C = 20 mA	h_{FE1}	40	160	
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), V_{CE} = 10 V dc, I_C = 2 mA	h_{FE2}	30		
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), V_{CE} = 10 V dc, I_C = 0.2 mA	h_{FE3}	10		
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3036	Bias condition D	I_{CB03}		100	µA dc
2N3439 2N3440		V_{CB} = 360 V dc V_{CB} = 250 V dc				
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 20 mA dc, pulsed (see 4.5.1)	h_{FE4}	15		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Pulse response:	3251	Test condition A				
Turn-on time		V_{CC} = 200 V dc, IC = 20 mA dc, I_{B1} = 2 mA dc, see figure 3	t_{on}		1	µs
Turn-off time		V_{CC} = 200 V dc, IC = 20 mA dc, I_{B1} = -I_{B2} = 2 mA dc, see figure 3	t_{off}		10	µs
Magnitude of common- emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc, I_C = 10 mA dc, f = 5 MHz	$\|h_{fe}\|$	3	15	
Open capacitance input open circuited	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		10	pF
Small-signal short- circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc, I_C = 5 mA, f = 1 kHz	h_{fe}	25		
Input capacitance (output open circuited)	3240	V_{CB} = 5 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		75	

See footnotes at end of table.

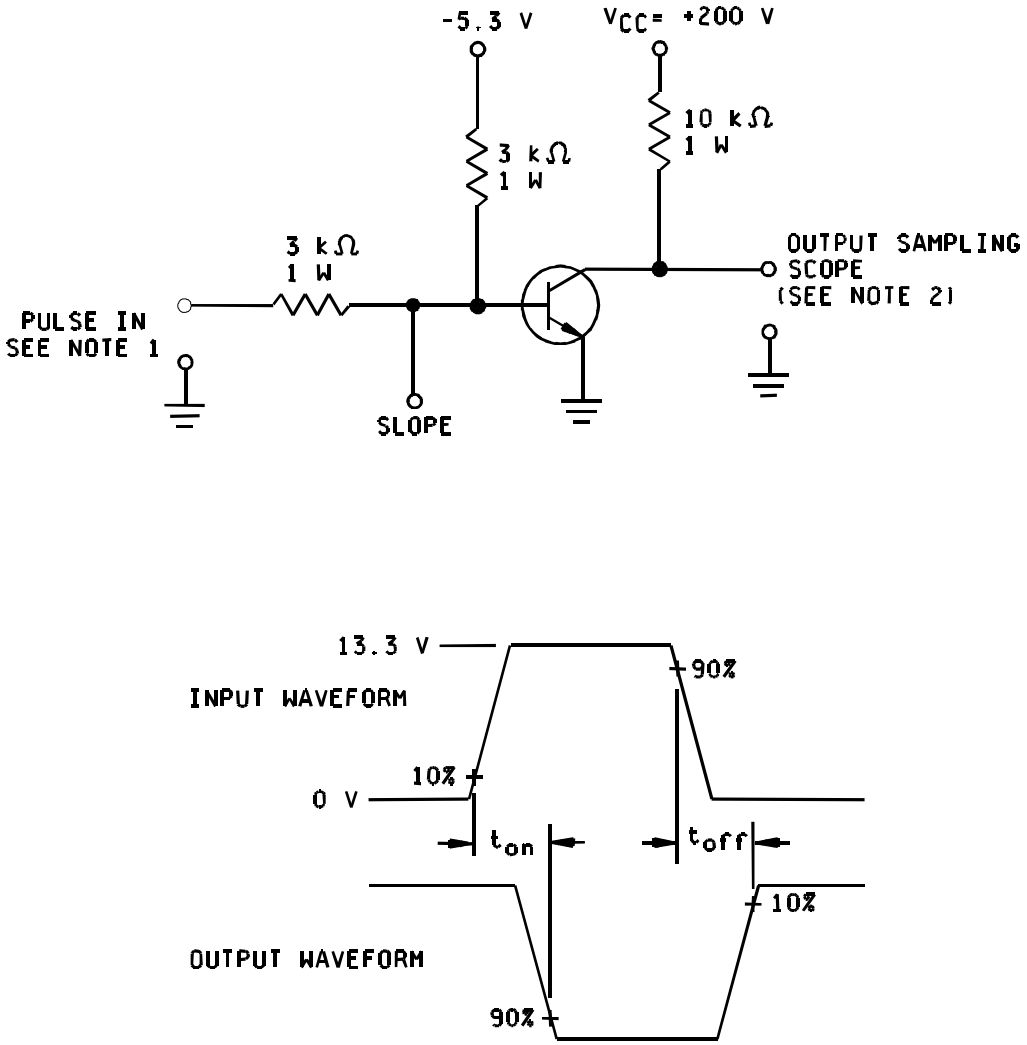
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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (continuous dc)	3051	(See figure 4) T_C = +25°C, 1 cycle, t = 1.0 s.				
Test 1		V_{CE} = 5 V dc, I_C = 1 A dc				
Test 2						
Only 2N3439		V_{CE} = 350 V dc, I_C = 14 mA dc				
Test 3						
Only 2N3440		V_{CE} = 250 V dc, I_C = 20 mA dc				
Electrical measurements		See table I, subgroup 2 and 4.5.3 herein.				
Breakdown voltage, collector to emitter	3011	I_C = 10 mA, R_{BB1} = 470 ohms V_{BB1} = 6 V, L = 25 mH minimum f = 30 to 60 Hz	$V_{BR(CEO)}$			V dc
2N3439				350		
2N3440				250		
Subgroup 7 4/						
Decap internal visual (design verification)	2075	n = 1 device, c = 0				

1/ For sampling plan, see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests.
3/ Separate samples may be used.
4/ Not required for JANS.
5/ Not required for laser marked devices.

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NOTES:

1. The rise time (t_r) and fall time (t_f) of applied pulse shall be ≤ 20 ns, duty cycle ≤ 2 percent, generator source impedance shall be 50 ohms, pulse width = 20 µs.
2. Output sampling oscilloscope: $Z_{IN} \geq 100$ kΩ, $C_{IN} \leq 50$ pF, and rise time ≤ 1.0 µs.

FIGURE 3. Pulse response test circuit.

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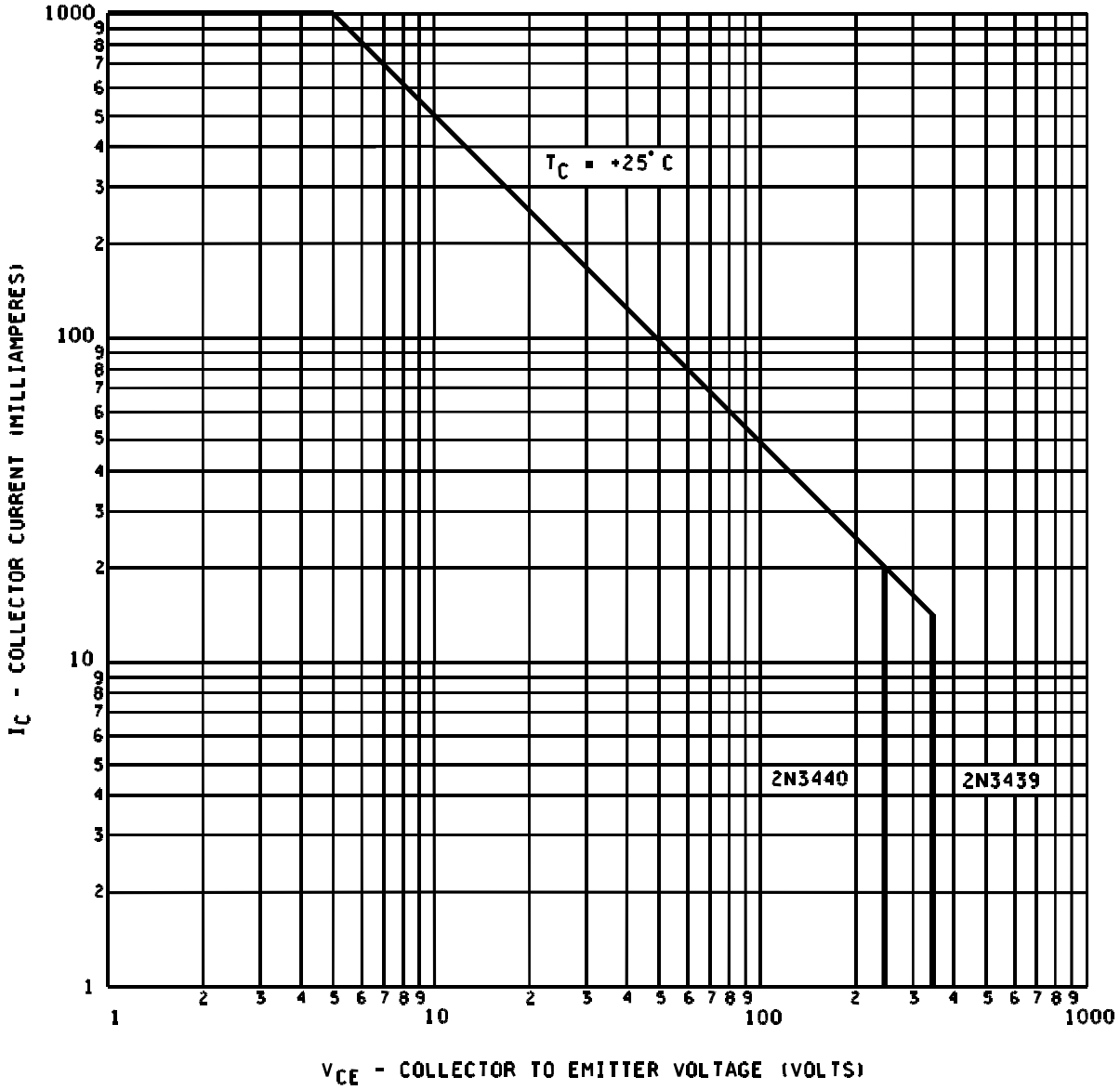


FIGURE 4. Maximum safe operating graph (continuous dc).

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-PRF-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Issue of DODISS to be cited in the solicitation (see 2.2.1).

b. The lead finish as specified (see 3.4.1).

c. Type designation and quality assurance level.

d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturer's List QML No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - MI, SM
Navy - AS, CG, MC
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5961-2061) ms

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/368D	**2. DOCUMENT DATE (YYMMDD)** 990630

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW POWER TYPES 2N3439, 2N3439L, 2N3440, and 2N3440L, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. Point of contact: Alan Barone	b. TELEPHONE Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan_barone@dscc.dla.mil
c. ADDRESS: Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-68880

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